Constructing a Layer 2 In-Path WLAN

Difficulty (out of 5 stars): ★★★★ Recommended study time: 3 hours

Contents

1 About This Lab

1.1 Task Description

Layer 2 refers to a topology where APs and ACs are connected within the same Layer 2 network segment, while Layer 3 refers to a topology where APs and ACs are connected across a Layer 3 network. In the Layer 2 networking, APs can go online in a plug-and-play manner by discovering an AC through Layer 2 broadcast or DHCP. In the Layer 3 networking, APs cannot directly discover an AC; instead, they must locate an AC dynamically through DHCP or DNS, or be configured with an AC's IP address.

In real-world deployments, an AC may manage dozens or even hundreds of APs, and the networking is usually complex. For example, on an enterprise network, APs may be installed in offices, meeting rooms, and reception rooms, while ACs are located in the enterprise's equipment room. In such cases, the ACs and APs are connected over a complex Layer 3 network. Therefore, large-scale deployments generally adopt the Layer 3 networking.

The objective of this task is to study the Layer 2 networking. Upon completing this task, you will gain a clear understanding of WLAN fundamentals and will be able to configure a WLAN using the Layer 2 networking.

1.2 Objectives

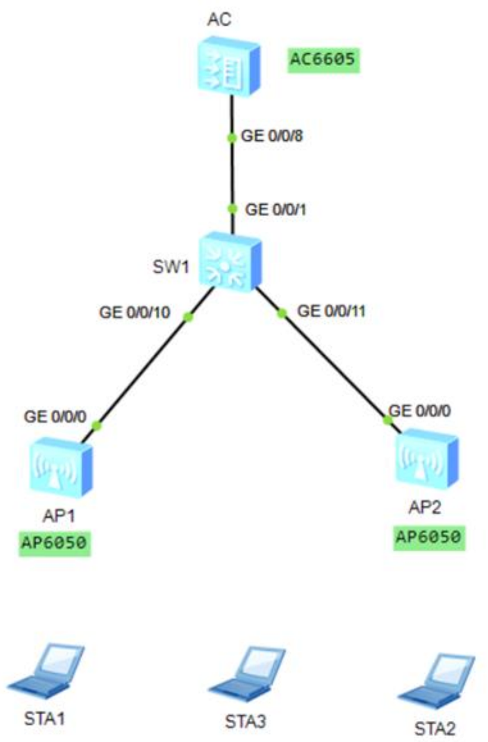
On completing this task, you will be able to:

1. Understand the commands for configuring a WLAN.
2. Understand the methods for configuring an AC.
3. Understand the configuration procedure for authenticating and onboarding APs.
4. Understand various WLAN configuration profiles.
5. Understand the basic process for WLAN configuration.
6. Understand the configuration roadmap of WLAN service parameters.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Layer 2 in-path WLAN networking topology



2.2 Initial Configuration

- Initial configuration of SW1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname SW1
[SW1]
```

- Initial configuration of the AC

```
<AC6605>system-view
Enter system view, return user view with Ctrl+Z.
[AC6605]sysname AC
[AC]
```

3 Task Implementation

3.1 Checking Basic Configurations

Omitted

3.2 Configuring Switches

1. Configure the access switch SW1. Add GE0/0/10 and GE0/0/11 to VLAN 10 (management VLAN, connected to APs) and set the PVID of the two interfaces to VLAN 10. Add GE0/0/1 to VLANs 10 to 13 (connected to the AC).

```
[SW1] vlan batch 10 to 13
Info: This operation may take a few seconds. Please wait for a moment...done.
[SW1]
[SW1] interface GigabitEthernet0/0/10
[SW1-GigabitEthernet0/0/10] port link-type trunk
[SW1-GigabitEthernet0/0/10] port trunk pvid vlan 10
[SW1-GigabitEthernet0/0/10] port trunk allow-pass vlan 10 to 13
[SW1-GigabitEthernet0/0/10] quit
[SW1]
[SW1] interface GigabitEthernet0/0/11
[SW1-GigabitEthernet0/0/11] port link-type trunk
[SW1-GigabitEthernet0/0/11] port trunk pvid vlan 10
[SW1-GigabitEthernet0/0/11] port trunk allow-pass vlan 10 to 13
[SW1-GigabitEthernet0/0/11] quit
[SW1]
[SW1] interface GigabitEthernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type trunk
[SW1-GigabitEthernet0/0/1] port trunk allow-pass vlan 10 to 13
[SW1-GigabitEthernet0/0/1] quit
[SW1]
```

2. Create the Loopback1 interface on SW1 and set its IP address to 101.101.101.101. Create VLANIF interfaces as the gateways for service VLANs.

```
[SW1] interface LoopBack 1
[SW1-LoopBack1] ip address 101.101.101.101 32
[SW1-LoopBack1] quit
[SW1] interface Vlanif 10
[SW1-Vlanif10] ip address 10.1.10.1 24
[SW1-Vlanif10] quit
[SW1]
[SW1] interface Vlanif 11
[SW1-Vlanif11] ip address 10.1.11.1 24
[SW1-Vlanif11] quit
[SW1]
[SW1] interface Vlanif 12
[SW1-Vlanif12] ip address 10.1.12.1 24
[SW1-Vlanif12] quit
[SW1]
[SW1] interface Vlanif 13
[SW1-Vlanif13] ip address 10.1.13.1 24
[SW1-Vlanif13] quit
[SW1]
```

3.3 Configuring Basic AC Information

1. Create VLANs 10 to 13.

```
[AC]vlan batch 10 to 13
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
[AC]
```

2. Configure GE0/0/8 to connect to SW1.

```
[AC]interface g0/0/8
[AC-GigabitEthernet0/0/8]port link-type trunk
[AC-GigabitEthernet0/0/8]port trunk allow-pass vlan 10 to 13
[AC-GigabitEthernet0/0/8]quit
[AC]
```

3. After the configuration is complete, run the **display port vlan** command to verify the configuration.

```
[AC]display port vlan
Port                    Link Type    PVID  Trunk VLAN List
-------------------------------------------------------------------------------
GigabitEthernet0/0/1    hybrid       1     -
GigabitEthernet0/0/2    hybrid       1     -
GigabitEthernet0/0/3    hybrid       1     -
GigabitEthernet0/0/4    hybrid       1     -
GigabitEthernet0/0/5    hybrid       1     -
GigabitEthernet0/0/6    hybrid       1     -
GigabitEthernet0/0/7    access       4090  -
GigabitEthernet0/0/8    trunk        1     1 10-13
```

4. Configure the IP addresses for Layer 3 VLANIF interfaces.

```
[AC]interface vlan 10
[AC-Vlanif10]ip address 10.1.10.100 24
[AC-Vlanif10]quit
[AC]
[AC]interface vlan 11
[AC-Vlanif11]ip address 10.1.11.100 24
[AC-Vlanif11]quit
[AC]
[AC]interface vlan 12
[AC-Vlanif12]ip address 10.1.12.100 24
[AC-Vlanif12]quit
[AC]
[AC]interface Vlanif 13
[AC-Vlanif13]ip address 10.1.13.100 24
[AC-Vlanif13]quit
[AC]
```

5. Verify the configuration.
   - Check whether the configured interfaces are Up.

```
[AC]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(E): E-Trunk down
The number of interface that is UP in Physical is 7
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 7
```

```
The number of interface that is DOWN in Protocol is 0

Interface                         IP Address/Mask      Physical    Protocol
NULL0                             unassigned           up          up(s)
Vlanif10                          10.1.10.100/24       up          up
Vlanif11                          10.1.11.100/24       up          up
Vlanif12                          10.1.12.100/24       up          up
Vlanif13                          10.1.13.100/24       up          up
```

- Check whether the route between the AC and the simulated public IP address of the Layer 3 switch is reachable. Note that at this point, pinging 101.101.101.101 (simulated public interface on the switch) will fail.

  On the AC, configure a static default route destined for the switch.

```
[AC]ip route-static 0.0.0.0 0.0.0.0 10.1.10.1
```

  Then, the IP address 101.101.101.101 can be pinged successfully.

```
[AC]ping 101.101.101.101
  PING 101.101.101.101: 56  data bytes, press CTRL_C to break
    Reply from 101.101.101.101: bytes=56 Sequence=1 ttl=254 time=1 ms
    Reply from 101.101.101.101: bytes=56 Sequence=2 ttl=254 time=1 ms
    Reply from 101.101.101.101: bytes=56 Sequence=3 ttl=254 time=1 ms
    Reply from 101.101.101.101: bytes=56 Sequence=4 ttl=254 time=1 ms
    Reply from 101.101.101.101: bytes=56 Sequence=5 ttl=254 time=1 ms

  --- 101.101.101.101 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/1/1 ms
```

3.4 Creating an AP Group

Create the AP group **ap-group1**.

```
[AC]wlan
[AC-wlan-view]ap-group name ap-group1
Info: This operation may take a few seconds. Please wait for a moment.done.
[AC-wlan-ap-group-ap-group1]quit
[AC-wlan-view]
```

3.5 Configuring AP Onboarding

1. Enable the DHCP service on the AC to assign IP addresses to STAs and APs. Configure Option 43 in the AP address pool.

```
[AC]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[AC]
[AC]ip pool ap
Info: It is successful to create an IP address pool.
[AC-ip-pool-ap]network 10.1.10.0 mask 24
[AC-ip-pool-ap]gateway-list 10.1.10.1
[AC-ip-pool-ap]option 43 sub-option 3 ascii 10.1.10.100
[AC-ip-pool-ap]quit
```

```
[AC]
[AC]ip pool employee
Info: It is successful to create an IP address pool.
[AC-ip-pool-employee]network 10.1.11.0 mask 24
[AC-ip-pool-employee]gateway-list 10.1.11.1
[AC-ip-pool-employee]quit
[AC]
[AC]ip pool voice
Info: It is successful to create an IP address pool.
[AC-ip-pool-voice]network 10.1.12.0 mask 24
[AC-ip-pool-voice]gateway-list 10.1.12.1
[AC-ip-pool-voice]quit
[AC]
[AC]ip pool guest
Info: It is successful to create an IP address pool.
[AC-ip-pool-guest]network 10.1.13.0 mask 24
[AC-ip-pool-guest]gateway-list 10.1.13.1
[AC-ip-pool-guest]quit
[AC]
```

2. Enable DHCP on each VLANIF interface of the AC.

```
[AC]interface Vlanif 10
[AC-Vlanif10]dhcp select global
[AC-Vlanif10]quit
[AC]
[AC]interface Vlanif 11
[AC-Vlanif11]dhcp select global
[AC-Vlanif11]quit
[AC]
[AC]interface Vlanif 12
[AC-Vlanif12]dhcp select global
[AC-Vlanif12]quit
[AC]
[AC]interface Vlanif 13
[AC-Vlanif13]dhcp select global
[AC-Vlanif13]quit
[AC]
```

3. Create the regulatory domain profile **domain***X* and configure the country code for the AC in the profile.

```
[AC]wlan
[AC-wlan-view]regulatory-domain-profile name domain1
[AC-wlan-regulate-domain-domain1]country-code CN
Info: The current country code is same with the input country code.
[AC-wlan-regulate-domain-domain1]quit
[AC-wlan-view]quit
[AC]
```

4. Configure the AC source interface.

```
[AC]capwap source interface vlanif 10
[AC]
```

5. Configure the AP authentication mode.

   Three AP authentication methods are supported. The default is MAC address authentication, which requires manually adding the AP list to the AC. If the authentication method has been changed, restore the default MAC address authentication as follows:

```
[AC]wlan
[AC-wlan-view]ap auth-mode mac-auth
[AC-wlan-view]
```

6. Manually import APs when they are offline.

   Import APs on the AC and add them to the AP group **ap-group1**. Manually add the APs based on their MAC addresses that are pasted on the underside of each AP, and name the APs **ap1** and **ap2**.

```
[AC1-wlan-view]ap-mac 00e0-fc8e-1060 ap-id 0
[AC1-wlan-ap-0]ap-group ap-group1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel, power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC1-wlan-ap-0]ap-name ap1
[AC1-wlan-ap-0]quit
[AC1-wlan-view]
[AC1-wlan-view]ap-mac 00e0-fcfb-6250 ap-id 1
[AC1-wlan-ap-1]ap-group ap-group1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel, power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC1-wlan-ap-1]ap-name ap2
[AC1-wlan-ap-1]quit
[AC1-wlan-view]quit
[AC1]
```

7. Verify AP onboarding.

   After an AP is added, its status changes from **fault** to **config**, and then to **normal**, and remains in **normal** state. If the status does not change to **normal** after a few minutes, check the VLAN, DHCP, and AP authentication configurations.

```
[AC]display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor    : normal                  [2]
--------------------------------------------------------------------------------
----------
ID     MAC                    Name Group        IP                Type                  State STA Uptime
--------------------------------------------------------------------------------
----------
0      00e0-fc8e-1060 ap1    ap-group1 10.1.10.172 AP6050DN            nor     0     1M:16S
1      00e0-fcfb-6250 ap2    ap-group1 10.1.10.178 AP6050DN            nor     0     25S
--------------------------------------------------------------------------------
----------
Total: 2
[AC]
```

3.6 Configuring WLAN Service Parameters

1. Configure SSID profiles.

   Create SSID profiles **employee1**, **voice1**, and **guest1**, and set the SSID names to **employee1**, **voice1**, and **guest1**, respectively.

```
[AC]wlan
[AC-wlan-view]ssid-profile name employee1
[AC-wlan-ssid-prof-employee1]ssid employee1
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-ssid-prof-employee1]quit
[AC-wlan-view]
[AC-wlan-view]ssid-profile name voice1
[AC-wlan-ssid-prof-voice1]ssid voice1
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-ssid-prof-voice1]quit
[AC-wlan-view]
[AC-wlan-view]ssid-profile name guest1
[AC-wlan-ssid-prof-guest1]ssid guest1
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-ssid-prof-guest1]quit
[AC-wlan-view]
```

2. Create VAP profiles.

   Create VAP profiles **employee1**, **voice1**, and **guest1**, set the data forwarding mode of **employee1** and **voice1** to direct forwarding and that of **guest1** to tunnel forwarding, configure service VLANs, and bind the SSID profiles to the VAP profiles.

```
[AC]wlan
[AC-wlan-view]vap-profile name employee1
[AC-wlan-vap-prof-employee1]forward-mode direct-forward
[AC-wlan-vap-prof-employee1]service-vlan vlan-id 11
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-vap-prof-employee1]ssid-profile employee1
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-vap-prof-employee1]quit
[AC-wlan-view]
[AC-wlan-view]vap-profile name voice1
[AC-wlan-vap-prof-voice1]forward-mode direct-forward
[AC-wlan-vap-prof-voice1]service-vlan vlan-id 12
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-vap-prof-voice1]ssid-profile voice1
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-vap-prof-voice1]quit
[AC-wlan-view]
[AC-wlan-view]vap-profile name guest1
[AC-wlan-vap-prof-guest1]forward-mode tunnel
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-vap-prof-guest1]service-vlan vlan-id 13
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-vap-prof-guest1]ssid-profile guest1
Info: This operation may take a few seconds, please wait.done.
[AC-wlan-vap-prof-guest1]quit
```

```
[AC-wlan-view]
```

3. Manage profiles.

   Bind the regulatory domain profile and VAP profiles to the AP group. Bind the VAP profiles **employee1**, **voice1**, and **guest1** to the AP group **ap-group1**, and set the VAP IDs in the VAP profiles to 1, 2, and 3, respectively. Apply the VAP profiles to radio 0 and radio 1 of APs.

```
[AC]wlan
[AC-wlan-view]ap-group name ap-group1
[AC-wlan-ap-group-ap-group1]vap-profile employee1 wlan 1 radio all
Info: This operation may take a few seconds, please wait...done.
[AC-wlan-ap-group-ap-group1]
[AC-wlan-ap-group-ap-group1]vap-profile voice1 wlan 2 radio all
Info: This operation may take a few seconds, please wait...done.
[AC-wlan-ap-group-ap-group1]
[AC-wlan-ap-group-ap-group1]vap-profile guest1 wlan 3 radio all
Info: This operation may take a few seconds, please wait...done.
[AC-wlan-ap-group-ap-group1]
[AC-wlan-ap-group-ap-group1]regulatory-domain-profile domain1
Warning: Modifying the country code will clear channel, power and antenna gain c
onfigurations of the radio and reset the AP. Continue?[Y/N]:
Error: Please choose 'YES' or 'NO' first before pressing 'Enter'. [Y/N]:y
[AC-wlan-ap-group-ap-group1]quit
[AC-wlan-view]quit
[AC]
```

4 Verification

4.1 Checking the VAP Status

WLAN service configurations are automatically delivered to APs. After the service configuration is complete, run the **display vap ssid employee***X*, **display vap ssid voice***X*, and **display vap ssid guest***X* commands. If the **Status** field is displayed as **ON**, the VAPs have been successfully created on the AP radios.

```
[AC]display vap ssid employee1
Info: This operation may take a few seconds, please wait.
WID : WLAN ID
---------------------------------------------------------------------------
AP ID AP name RfID WID   BSSID            Status   Auth type   STA    SSID
---------------------------------------------------------------------------
0      ap1      0     1    00E0-FC8E-1060 ON       Open         0      employee1
0      ap1      1     1    00E0-FC8E-1070 ON       Open         0      employee1
1      ap2      0     1    00E0-FCFB-6250 ON       Open         0      employee1
1      ap2      1     1    00E0-FCFB-6260 ON       Open         0      employee1
---------------------------------------------------------------------------
Total: 4
[AC]
[AC]display vap ssid voice1
Info: This operation may take a few seconds, please wait.
WID : WLAN ID
---------------------------------------------------------------------------
AP ID AP name RfID WID   BSSID            Status   Auth type   STA    SSID
```

```
--------------------------------------------------------------------------
0       ap1       0      2      00E0-FC8E-1061 ON        Open          0         voice1
0       ap1       1      2      00E0-FC8E-1071 ON        Open          0         voice1
1       ap2       0      2      00E0-FCFB-6251 ON        Open          0         voice1
1       ap2       1      2      00E0-FCFB-6261 ON        Open          0         voice1
--------------------------------------------------------------------------
Total: 4
[AC]
[AC]display vap ssid guest1
Info: This operation may take a few seconds, please wait.
WID : WLAN ID
--------------------------------------------------------------------------
AP ID AP name RfID WID   BSSID              Status   Auth type   STA     SSID
--------------------------------------------------------------------------
0       ap1       0      3      00E0-FC8E-1062 ON        Open          0         guest1
0       ap1       1      3      00E0-FC8E-1072 ON        Open          0         guest1
1       ap2       0      3      00E0-FCFB-6252 ON        Open          0         guest1
1       ap2       1      3      00E0-FCFB-6262 ON        Open          0         guest1
--------------------------------------------------------------------------
Total: 4
[AC]
```

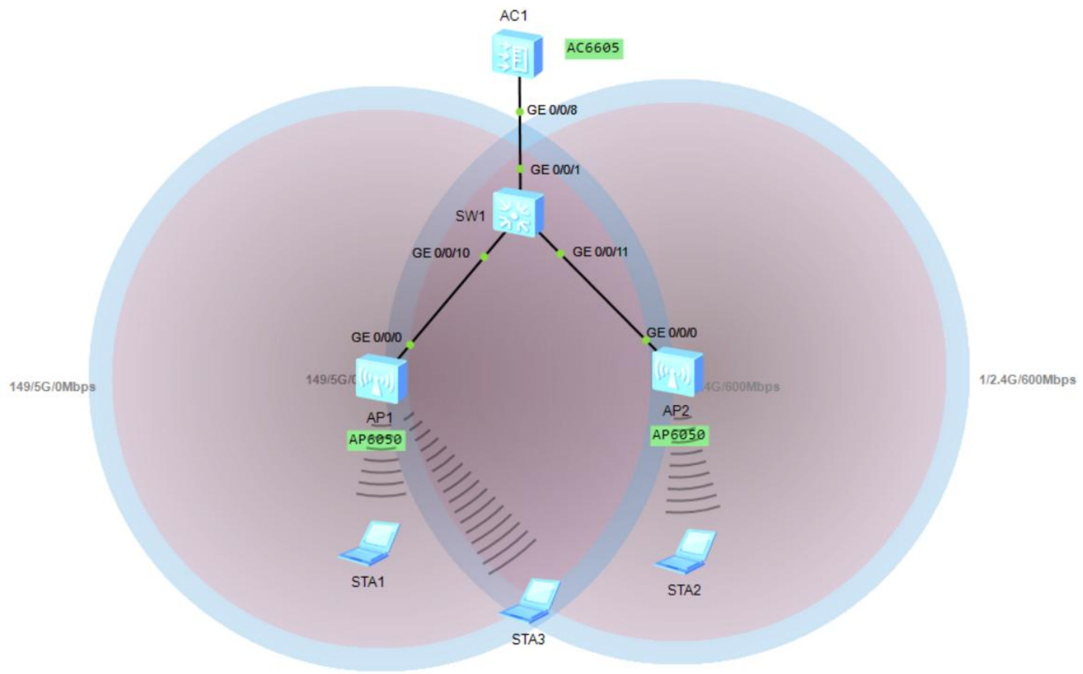
4.2 Connecting STAs

After the configuration is complete, the APs broadcast the **employee1**, **voice1**, and **guest1** signals, and the authentication mode is open.

After associating a STA with one of the wireless signals, you can run the **display station all** command to view information about the associated STA. Figure 4-1 shows the STA connections.

```
[AC]display station all
Rf/WLAN: Radio ID/WLAN ID
Rx/Tx: link receive rate/link transmit rate(Mbps)
-------------------------------------------------------------------------------
------------------
STA MAC              AP ID Ap name   Rf/WLAN   Band   Type   Rx/Tx        RSSI   VLAN   IP a
ddress   SSID
-------------------------------------------------------------------------------
------------------
5489-98af-58c2    1        ap2         0/3         2.4G   -        -/-            -       13       10.1
.13.198 guest1
5489-98dd-164b    0        ap1         0/2         2.4G   -        -/-            -       12       10.1
.12.21    voice1
5489-98ee-6ae0    0        ap1         0/1         2.4G   -        -/-            -       11       10.1
.11.231 employee1
-------------------------------------------------------------------------------
------------------
Total: 3 2.4G: 3 5G: 0
[AC]
```

Figure 4-1 STA connections



4.3 Testing Connectivity

On STAs, ping the simulated public IP address 101.101.101.101 of the switch.

- STA connected to the SSID **employee1**

```
STA>ipconfig

Link local IPv6 address...........: ::
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 10.1.11.231
Subnet mask.......................: 255.255.255.0
Gateway...........................: 10.1.11.1
Physical address..................: 54-89-98-EE-6A-E0
DNS server........................:

STA>ping 101.101.101.101 -c 2

Ping 101.101.101.101: 32 data bytes, Press Ctrl_C to break
From 101.101.101.101: bytes=32 seq=1 ttl=255 time=110 ms
From 101.101.101.101: bytes=32 seq=2 ttl=255 time=125 ms

--- 101.101.101.101 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 110/117/125 ms
```

```
STA>
```

- STA connected to the SSID **guest1**

```
STA>ipconfig

Link local IPv6 address...........: ::
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 10.1.13.198
Subnet mask.......................: 255.255.255.0
Gateway...........................: 10.1.13.1
Physical address..................: 54-89-98-AF-58-C2
DNS server........................:

STA>ping 101.101.101.101 -c 2

Ping 101.101.101.101: 32 data bytes, Press Ctrl_C to break
From 101.101.101.101: bytes=32 seq=1 ttl=255 time=140 ms
From 101.101.101.101: bytes=32 seq=2 ttl=255 time=141 ms

--- 101.101.101.101 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 140/140/141 ms

STA>
```

- STA connected to the SSID **voice1**

```
STA>ipconfig

Link local IPv6 address...........: ::
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 10.1.12.21
Subnet mask.......................: 255.255.255.0
Gateway...........................: 10.1.12.1
Physical address..................: 54-89-98-DD-16-4B
DNS server........................:

STA>ping 101.101.101.101 -c 2

Ping 101.101.101.101: 32 data bytes, Press Ctrl_C to break
From 101.101.101.101: bytes=32 seq=1 ttl=255 time=204 ms
From 101.101.101.101: bytes=32 seq=2 ttl=255 time=188 ms

--- 101.101.101.101 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
```

```
  round-trip min/avg/max = 188/196/204 ms

STA>
```

5 Quiz

1. What functions can CAPWAP tunnels provide?

   Maintains the running status of the AC and APs.

   Allows the AC to manage APs and deliver configurations to APs.

   Transmits service data to the AC for centralized forwarding.

2. How does an AP discover an AC in static mode?

   An AC IP address list is preconfigured on the AP. When the AP goes online, it unicasts a Discovery Request packet to each AC whose IP address is specified in the preconfigured AC IP address list. After receiving the Discovery Request message, the ACs send Discovery Response messages to the AP. The AP then selects an AC to establish a CAPWAP tunnel according to the received Discovery Response messages.

3. How does an AP dynamically discover an AC?

   An AP can dynamically discover an AC in DHCP, DNS, or broadcast mode. The DHCP and broadcast modes are described in this lab guide.